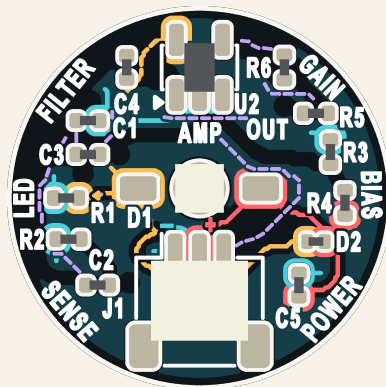


Meet your circuit.

Match the labels. Follow the connections.



U1 is on the other PCB face.

k = thousand M = million uF = microfarads

● + supply

● Ground (-)

● Signal

● PROT / LED

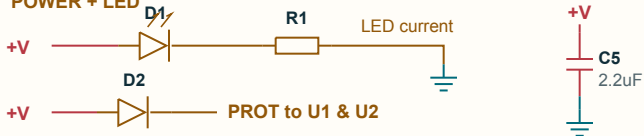
TL;DR Power in. Filter + amplify. Signal out.

PARTS LIST / BOM

REF	PART / VALUE
R1	LED limit
R2	12k ohm
R3, R4	100k ohm
R5	10k ohm
R6	3.3M ohm
C1-C3	4.7 uF
C4, C5	2.2 uF
D1	Green LED
D2	Schottky diode
U1	APDS-9008
U2	MCP6001
J1	3-pin JST SH

Same parts. Different view.

1 POWER + LED



2 INPUT

PROT

U1
LIGHT
SENSOR



$C2$

4.7uF



3 FILTER

$C3$

4.7uF



$R5$

10k

$C4$

2.2uF

VREF

4 AMPLIFY

$R6$

3.3M



U2 power:

PROT / GND

BIAS



$R3 = R4 = 100k$

Sets the signal midpoint.

R = resistor

C = capacitor

U = chip

J1: 1 GND (-) 2 +V 3 OUT

Matching labels connect.

PROT = protected supply after D2